BEX6510 Notes

Matrix Algebra

Linear Models

Notation

The Multivariate Linear Model: The general linear model is written

$$\underset{(T\times 1)}{\mathbf{y}} = \underset{(T\times k)}{\mathbf{X}}\underset{(k\times 1)}{\boldsymbol{\beta}} + \underset{(T\times 1)}{\boldsymbol{\varepsilon}}$$

where now

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ . \\ . \\ y_T \end{bmatrix}, \mathbf{X} = \begin{bmatrix} x_{11} & x_{12} & . & . & x_{1k} \\ x_{21} & x_{22} & . & . & x_{2k} \\ . & & . & & . \\ . & & & . & . \\ x_{T1} & . & . & . & x_{Tk} \end{bmatrix}, \boldsymbol{\beta} = \begin{bmatrix} \beta_1 \\ \beta_2 \\ . \\ \beta_k \end{bmatrix} \quad \text{and} \quad \underset{(T\times 1)}{\boldsymbol{\varepsilon}} = \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ . \\ . \\ \varepsilon_T \end{bmatrix}.$$

This is derived from the t'th observation, which we express as

$$\begin{aligned} y_t &= x_{t1}\beta_1 + x_{t2}\beta_2 + ... + x_{tk}\beta_k + \varepsilon_t, \quad t = 1, \ldots, T, \quad \text{or} \\ y_t &= \mathbf{x}_t'\boldsymbol{\beta} + \varepsilon_t, \quad t = 1, \ldots, T, \quad \text{where} \quad \mathbf{x}_t' = (x_{t1}, x_{t2}, ..., x_{tk}) \ . \end{aligned}$$

Assumptions/regularity conditions

$$
\begin{array}{lll}
(A1) & : & E\left[\varepsilon_t | X_t\right] = 0 \qquad \text{(Zero Mean Disturbance)} \\
(A2) & : & Var\left[\varepsilon_t | X_t\right] = \sigma_\varepsilon^2 \qquad \text{(Homoscedasticity – Constant Variance)} \\
(A3) & : & Cov\left[\varepsilon_t \varepsilon_s | X_t X_s\right] = 0 \text{ for all } t, s \qquad \text{(Non-autocorrelated)} \\
(A4) & : & Cov\left[\varepsilon_t X_s\right] = 0 \text{ for all } t, s \qquad \text{(Exogenous Regressor)} \\
(A5) & : & \varepsilon_t \sim N\left(0, \sigma_\varepsilon^2\right) \qquad \text{(Gaussian / Normal Distribution)} \\
(A6) & : & \sum_{t=1}^{T} \left(x_t - \overline{x}\right)^2 > 0 \qquad \text{(Identification)}
\end{array}
$$

Useful matrix algebra identity

$$
\mathbf{X'X} = \underset{(2\times T)}{\begin{bmatrix} 1 & . & . & 1 \\ x_1 & . & . & x_T \end{bmatrix}} \underset{(T\times 2)}{\begin{bmatrix} 1 & x_1 \\ . & . \\ . & . \\ 1 & x_T \end{bmatrix}} = \underset{(2\times 2)}{\begin{bmatrix} T & \sum x_t \\ \sum x_t & \sum x_t^2 \end{bmatrix}}
$$

$$
\mathbf{X'y} = \underset{(2\times T)}{\begin{bmatrix} 1 & . & . & 1 \\ x_1 & . & . & x_T \end{bmatrix}} \underset{(T\times 1)}{\begin{bmatrix} y_1 \\ . \\ . \\ y_T \end{bmatrix}} = \underset{(2\times 1)}{\begin{bmatrix} \sum y_t \\ \sum x_t y_t \end{bmatrix}}
$$

FWL Theorem

Theorem 1.8 *If the regressors are partitioned as in* (1.1) *then:*

1. *If* $\mathbf{b}_1$ *and* $\mathbf{b}_2$ *are the OLS regression coefficients in the regression*

$$\mathbf{y} = \mathbf{X}\mathbf{b} + \mathbf{e} = \mathbf{X}_1\mathbf{b}_1 + \mathbf{X}_2\mathbf{b}_2 + \mathbf{e}$$

then

$$\mathbf{b}_2 = \left(\mathbf{X}_2'\mathbf{R}_1\mathbf{X}_2\right)^{-1}\mathbf{X}_2'\mathbf{R}_1\mathbf{y}$$

and

$$\mathbf{b}_1 = \left(\mathbf{X}_1'\mathbf{R}_2\mathbf{X}_1\right)^{-1}\mathbf{X}_1'\mathbf{R}_2\mathbf{y}$$

where $\mathbf{R}_i = \mathbf{I}_T - \mathbf{P}_i$ *and* $\mathbf{P}_i = \mathbf{X}_i\left(\mathbf{X}_i'\mathbf{X}_i\right)^{-1}\mathbf{X}_i'$, $i = 1, 2$.

2. *The residual*

$$\begin{aligned} \mathbf{e} &= \mathbf{y} - \mathbf{X}_1\mathbf{b}_1 - \mathbf{X}_2\mathbf{b}_2 \\ &= \mathbf{R}_1\left(\mathbf{y} - \mathbf{X}_2\mathbf{b}_2\right) \\ &= \mathbf{R}_2\left(\mathbf{y} - \mathbf{X}_1\mathbf{b}_1\right). \end{aligned}$$

Note that $\mathbf{P}_i' = \mathbf{P}_i$, $\mathbf{P}_i\mathbf{P}_i = \mathbf{P}_i$ and $\mathbf{R}_i' = \mathbf{R}_i$, $\mathbf{R}_i\mathbf{R}_i = \mathbf{R}_i$, and

$$\begin{aligned} \mathbf{P}_i\mathbf{y} &= \mathbf{X}_i\left(\mathbf{X}_i'\mathbf{X}_i\right)\mathbf{X}_i\mathbf{y} = \widehat{\mathbf{y}}_i && \text{Projection/Prediction} \\ \mathbf{R}_i\mathbf{y} &= \mathbf{y} - \widehat{\mathbf{y}}_i = \mathbf{e}_i && \text{Residual} \end{aligned}$$

obtained using *only* the regressors in $\mathbf{X}_i$.

Statistical Properties of OLS

Proposition 1.10 $Var\left[\mathbf{b}|\mathbf{X}\right] = \sigma_{\varepsilon}^{2}\left(\mathbf{X}'\mathbf{X}\right)^{-1}$.

Proof. By definition

$$\begin{aligned}
Var\left[\mathbf{b}|\mathbf{X}\right] &= E\left[\left(\mathbf{b}-\boldsymbol{\beta}\right)\left(\mathbf{b}-\boldsymbol{\beta}\right)'|\mathbf{X}\right] \\
&= E\left[\left(\left(\mathbf{X}'\mathbf{X}\right)^{-1}\mathbf{X}'\varepsilon\right)\left(\left(\mathbf{X}'\mathbf{X}\right)^{-1}\mathbf{X}'\varepsilon\right)'|\mathbf{X}\right] \\
&= \left(\mathbf{X}'\mathbf{X}\right)^{-1}\mathbf{X}'E\left[\varepsilon\varepsilon'|\mathbf{X}\right]\mathbf{X}\left(\mathbf{X}'\mathbf{X}\right)^{-1} \\
&= \left(\mathbf{X}'\mathbf{X}\right)^{-1}\mathbf{X}'\sigma_{\varepsilon}^{2}\mathbf{I}_{T}\mathbf{X}\left(\mathbf{X}'\mathbf{X}\right)^{-1} \\
&= \sigma_{\varepsilon}^{2}\left(\mathbf{X}'\mathbf{X}\right)^{-1}
\end{aligned}$$

Proof of variance of residuals

$$\begin{aligned}
E\left[\mathbf{e}'\mathbf{e}|\mathbf{X}\right] &= E\left[\varepsilon'\mathbf{R}\varepsilon|\mathbf{X}\right] \\
&= E\left[tr\left(\varepsilon'\mathbf{R}\varepsilon\right)|\mathbf{X}\right] \\
&= E\left[tr\left(\mathbf{R}\varepsilon\varepsilon'\right)|\mathbf{X}\right] \\
&= tr\left(\mathbf{R}E\left[\varepsilon\varepsilon'|\mathbf{X}\right]\right) \\
&= \sigma_{\varepsilon}^{2}tr(\mathbf{R}) \\
&= \sigma_{\varepsilon}^{2}tr(\mathbf{I}_{T}-\mathbf{P}) \\
&= \sigma_{\varepsilon}^{2}\left[tr\left(\mathbf{I}_{T}\right)-tr\left(\mathbf{X}\left(\mathbf{X}'\mathbf{X}\right)^{-1}\mathbf{X}'\right)\right] \\
&= \sigma_{\varepsilon}^{2}\left[T-tr\left(\left(\mathbf{X}'\mathbf{X}\right)^{-1}\mathbf{X}'\mathbf{X}\right)\right] \\
&= \sigma_{\varepsilon}^{2}\left[T-tr\left(\mathbf{I}_{k}\right)\right] \\
&= \sigma_{\varepsilon}^{2}\left(T-k\right)
\end{aligned}$$

Inference

The very basic case

Proposition 4.1 *The quantity* $\mathbf{l}'\mathbf{b} \sim N\left(\mathbf{l}'\beta, \sigma_\varepsilon^2 \mathbf{l}'\left(\mathbf{X}'\mathbf{X}\right)^{-1}\mathbf{l}\right)$ *or equivalently*

$$\frac{\mathbf{l}'\left(\mathbf{b}-\beta\right)}{\sigma_\varepsilon\sqrt{\mathbf{l}'\left(\mathbf{X}'\mathbf{X}\right)^{-1}\mathbf{l}}} \sim N\left(0,1\right)$$

Theorem 4.2 *If* $s^2 = \mathbf{e}'\mathbf{e}/\left(T-k\right)$ *then*

$$\frac{\mathbf{l}'\left(\mathbf{b}-\beta\right)}{s\sqrt{\mathbf{l}'\left(\mathbf{X}'\mathbf{X}\right)^{-1}\mathbf{l}}} \sim t_{(T-k)}$$

a "Student t distribution with $(T-k)$ *degrees of freedom."*

More generally

Proposition 4.4 *The variable* $\mathbf{L}\mathbf{b} \sim N\left(\mathbf{L}\beta, \sigma_\varepsilon^2 \mathbf{L}\left(\mathbf{X}'\mathbf{X}\right)^{-1}\mathbf{L}'\right)$.

Lemma 4.1 *The quadratic form*

$$Q = \frac{\left(\mathbf{b}-\beta\right)'\mathbf{L}'\left(\mathbf{L}\left(\mathbf{X}'\mathbf{X}\right)^{-1}\mathbf{L}'\right)^{-1}\mathbf{L}\left(\mathbf{b}-\beta\right)}{\sigma_\varepsilon^2} \sim \chi^2\left(J\right)$$

Recognize that:

1. $\left(T-k\right)s^2/\sigma_\varepsilon^2 \sim \chi^2_{(T-k)}$
2. The variables s^2 and Q are independently distributed
3. Ratio of independently χ^2 random variables, divided by their respective d.o.f.,

$$\frac{\chi^2\left(J\right)/J}{\chi^2\left(T-k\right)^2/(T-k)}$$

$$\begin{aligned} &\sim F\left(\text{d.o.f numerator, d.o.f denominator}\right) \\ &\sim F\left(J, T-k\right) \end{aligned}$$

Restricted Least Squares (Lagrange Approach)

Proof. The constrained solution is obtained by minimizing the Lagrangian function

$$\Lambda = (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})' (\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) + 2 (\mathbf{L}\boldsymbol{\beta} - \mathbf{r})' \boldsymbol{\lambda}$$

Note the choice of choosing the lagrange multiplier to has a constant in front to simplify things!

and by definition of $\mathbf{b}^*$ and the associated $\boldsymbol{\lambda}^*$

$$\begin{aligned} \frac{\partial \Lambda}{\partial \boldsymbol{\beta}}|_{\beta=b^*} &= -2\mathbf{X}'\mathbf{y} + 2\mathbf{X}'\mathbf{X}\mathbf{b}^* + 2\mathbf{L}'\boldsymbol{\lambda}^* = \mathbf{0} && (i) \\ \frac{\partial \Lambda}{\partial \boldsymbol{\lambda}}|_{\lambda=\lambda^*} &= 2 (\mathbf{L}\mathbf{b}^* - \mathbf{r}) = \mathbf{0} && (ii) \end{aligned}$$

From (i)

$$\begin{aligned} (\mathbf{X}'\mathbf{X})\, \mathbf{b}^* &= \mathbf{X}'\mathbf{y} - \mathbf{L}'\boldsymbol{\lambda}^* \\ \mathbf{b}^* &= \mathbf{b} - (\mathbf{X}'\mathbf{X})^{-1}\mathbf{L}'\boldsymbol{\lambda}^* \end{aligned}$$

Therefore

$$\mathbf{L}\mathbf{b}^* = \mathbf{L}\mathbf{b} - \mathbf{L} (\mathbf{X}'\mathbf{X})^{-1}\mathbf{L}'\boldsymbol{\lambda}^* = \mathbf{r}$$

from (ii). So

$$\begin{aligned} \boldsymbol{\lambda}^* &= \left(\mathbf{L} (\mathbf{X}'\mathbf{X})^{-1}\mathbf{L}'\right)^{-1} (\mathbf{L}\mathbf{b} - \mathbf{L}\mathbf{b}^*) \\ &= \left(\mathbf{L} (\mathbf{X}'\mathbf{X})^{-1}\mathbf{L}'\right)^{-1} (\mathbf{L}\mathbf{b} - \mathbf{r}) \end{aligned}$$

and substituting back into the previous expression for $\mathbf{b}^*$ gives

$$\mathbf{b}^* = \mathbf{b} + (\mathbf{X}'\mathbf{X})^{-1} \mathbf{L}' \left(\mathbf{L} (\mathbf{X}'\mathbf{X})^{-1} \mathbf{L}'\right)^{-1} (\mathbf{r} - \mathbf{L}\mathbf{b}) ,$$

as required. ■

Chow test

Normal derivation

Consider

$$(M) : \mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \varepsilon$$

and let

$$\mathbf{y} = \begin{bmatrix} \mathbf{y}_{(1)} \\ \mathbf{y}_{(2)} \end{bmatrix} \quad \mathbf{X} = \begin{bmatrix} \mathbf{X}_{(1)} \\ \mathbf{X}_{(2)} \end{bmatrix} \quad \text{and} \quad \varepsilon = \begin{bmatrix} \varepsilon_{(1)} \\ \varepsilon_{(2)} \end{bmatrix}$$

indicating a partition of the data into two groups/subsets. An implicit assumption is that the regression coefficient is the same in the two groups. The "unrestricted" model states that

$$\begin{aligned} \mathbf{y}_{(1)} &= \mathbf{X}_{(1)}\boldsymbol{\beta}_{(1)} + \varepsilon_{(1)} \\ \mathbf{y}_{(2)} &= \mathbf{X}_{(2)}\boldsymbol{\beta}_{(2)} + \varepsilon_{(2)} \end{aligned}$$

The test statistic becomes

$$\lambda = \frac{(T-2k)}{k}\left[\frac{\mathbf{e}'\mathbf{e}}{\mathbf{e}'_{(1)}\mathbf{e}_{(1)} + \mathbf{e}'_{(2)}\mathbf{e}_{(2)}} - 1\right]$$

Under H_0 we have $\lambda \sim F(k, T-2k)$ and

$$\text{If } \lambda \leq F_{(1-\alpha)}(J, T-k) \text{ do not reject } H_0$$
$$\text{If } \lambda > F_{(1-\alpha)}(J, T-k) \text{ reject } H_0$$

N.B.

$$\begin{aligned} RSS_{(1)} &= \mathbf{e}'_{(1)}\mathbf{e}_{(1)} \text{ residual sum of squares using } T_1 \text{ observations group 1.} \\ RSS_{(2)} &= \mathbf{e}'_{(2)}\mathbf{e}_{(2)} \text{ residual sum of squares using } T_2 \text{ observations group 2.} \end{aligned}$$

Thus $RSS_U = RSS_{(1)} + RSS_{(2)}$. Also

$$RSS_{(1+2)} = \mathbf{e}'\mathbf{e} \text{ residual sum of squares using all } T = T_1 + T_2 \text{ observations.}$$

This is the restricted residual sum of squares $RSS_{(1+2)} = RSS_R$. Therefore

$$\lambda = \frac{(T_1 + T_2 - 2k)}{k}\left[\frac{RSS_{(1+2)}}{RSS_{(1)} + RSS_{(2)}} - 1\right]$$

Alternative derivation when one of the sub samples has less observations than regressors:

$$
\begin{aligned}
\mathbf{y}_{(1)} &= \mathbf{X}_{(1)}\boldsymbol{\beta}_{(1)} + \boldsymbol{\varepsilon}_{(1)} \\
\mathbf{y}_{(2)} &= \mathbf{X}_{(2)}\boldsymbol{\beta}_{(2)} + \mathbf{X}_{(2)}\left(\boldsymbol{\beta}_{(2)} - \boldsymbol{\beta}_{(1)}\right) + \boldsymbol{\varepsilon}_{(2)}
\end{aligned}
$$

or

$$
\mathbf{y} = \begin{bmatrix} \mathbf{X}_{(1)} & \mathbf{0} \\ \mathbf{X}_{(2)} & \mathbf{X}_{(2)} \end{bmatrix} \begin{bmatrix} \boldsymbol{\beta}_{(1)} \\ \Delta\boldsymbol{\beta} \end{bmatrix} + \boldsymbol{\varepsilon}
$$

where $\Delta\boldsymbol{\beta} = \boldsymbol{\beta}_{(2)} - \boldsymbol{\beta}_{(1)}$. A test of $H_0 : \boldsymbol{\beta}_{(2)} = \boldsymbol{\beta}_{(1)}$ is therefore equivalent to testing the joint significance of the last k regressors, namely $H_0 : \Delta\boldsymbol{\beta} = \boldsymbol{\beta}_{(2)} - \boldsymbol{\beta}_{(1)} = \mathbf{0}$.

Now suppose that $T_1 > k$ and $T_2 \leq k$. Then we still can express the "unrestricted" model as

$$
\begin{aligned}
(M_U) : \begin{bmatrix} \mathbf{y}_{(1)} \\ \mathbf{y}_{(2)} \end{bmatrix} &= \begin{bmatrix} \mathbf{X}_{(1)} & \mathbf{0} \\ \mathbf{X}_{(2)} & \mathbf{X}_{(2)} \end{bmatrix} \begin{bmatrix} \boldsymbol{\beta}_{(1)} \\ \boldsymbol{\beta}_{(2)} - \boldsymbol{\beta}_{(1)} \end{bmatrix} + \begin{bmatrix} \boldsymbol{\varepsilon}_{(1)} \\ \boldsymbol{\varepsilon}_{(2)} \end{bmatrix} \\
&= \begin{bmatrix} \mathbf{X}_{(1)} & \mathbf{0} \\ \mathbf{X}_{(2)} & \mathbf{I} \end{bmatrix} \begin{bmatrix} \boldsymbol{\beta} \\ \boldsymbol{\delta} \end{bmatrix} + \begin{bmatrix} \boldsymbol{\varepsilon}_{(1)} \\ \boldsymbol{\varepsilon}_{(2)} \end{bmatrix}
\end{aligned}
$$

where $\boldsymbol{\beta} = \boldsymbol{\beta}_{(1)}$ and $\boldsymbol{\delta} = \mathbf{X}_{(2)}\Delta\boldsymbol{\beta} = \mathbf{X}_{(2)}\left(\boldsymbol{\beta}_{(2)} - \boldsymbol{\beta}_{(1)}\right)$

But $H_0 : \Delta\boldsymbol{\beta} = \boldsymbol{\beta}_{(2)} - \boldsymbol{\beta}_{(1)} = \mathbf{0}$ implies that $\boldsymbol{\delta} = \mathbf{0}$ and the "restricted" model becomes

$$
\begin{aligned}
(M_R) : \begin{bmatrix} \mathbf{y}_{(1)} \\ \mathbf{y}_{(2)} \end{bmatrix} &= \begin{bmatrix} \mathbf{X}_{(1)}\boldsymbol{\beta} \\ \mathbf{X}_{(2)}\boldsymbol{\beta} \end{bmatrix} + \begin{bmatrix} \boldsymbol{\varepsilon}_{(1)} \\ \boldsymbol{\varepsilon}_{(2)} \end{bmatrix} \\
\text{or} \quad \mathbf{y} &= \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}
\end{aligned}
$$

where $\boldsymbol{\beta} = \boldsymbol{\beta}_{(1)} = \boldsymbol{\beta}_{(2)}$. Thus

$$
RSS_U = \mathbf{y}'\mathbf{y} - \mathbf{y}'\mathbf{Z}\left(\mathbf{Z}'\mathbf{Z}\right)^{-1}\mathbf{Z}'\mathbf{y}
$$

which is equal to the residual sum of squares of a regression of

$$
\mathbf{y} = \begin{bmatrix} \mathbf{y}_{(1)} \\ \mathbf{y}_{(2)} \end{bmatrix} \text{ on } \mathbf{Z} = \begin{bmatrix} \mathbf{X}_{(1)} & \mathbf{0} \\ \mathbf{X}_{(2)} & \mathbf{I} \end{bmatrix}.
$$

This is an F test with (T - T2 - k) = (T1 - k) dof and T2 indicator variables (restrictions).

The test statistic in this case becomes

$$\lambda = \frac{(T_1 - k)}{T_2}\left[\frac{RSS_{(1+2)}}{RSS_{(1)}} - 1\right] \sim F(T_2, T_1 - k)$$

Likelihood

KL divergence. Note that the relative orders of the thetas are reversed in the fraction.

Definition 5.4 *The Kullback-Leibler divergence*

$$K(\boldsymbol{\theta} : \boldsymbol{\theta}_0) = E\left[\log\left\{\frac{f(z : \boldsymbol{\theta}_0)}{f(z : \boldsymbol{\theta})}\right\}\right] = \int_{S_Z} \log\left\{\frac{f(z : \boldsymbol{\theta}_0)}{f(z : \boldsymbol{\theta})}\right\} f(z : \boldsymbol{\theta}_0)\, dz$$

where $f(z : \boldsymbol{\theta})$ is the p.d.f. of the random variable Z with sample space S_Z.

Thus $K(\boldsymbol{\theta} : \boldsymbol{\theta}_0)$ is the expected value of the log likelihood ratio evaluated for a single data point. If $z_1, \ldots, z_n$ are i.i.d. with p.d.f. $f(z : \boldsymbol{\theta})$ then

$$K_n(\boldsymbol{\theta} : \boldsymbol{\theta}_0) = E\left[\log\left\{\frac{\prod_{i=1}^n f(z_i : \boldsymbol{\theta}_0)}{\prod_{i=1}^n f(z_i : \boldsymbol{\theta})}\right\}\right] = \int_{S_{Z_n}} \cdots \int_{S_{Z_1}} \log\left\{\frac{\prod_{i=1}^n f(z_i : \boldsymbol{\theta}_0)}{\prod_{i=1}^n f(z_i : \boldsymbol{\theta})}\right\} \prod_{i=1}^n f(z_i : \boldsymbol{\theta}_0)\, dz_i\,.$$

This proof is straightforward, but tedious. Focus on a case of n = 2. Just expand out the multiplication, etc.

The multiple n-fold integral on the right hand side simplifies to

$$\sum_{i=1}^n \int_{S_{Z_i}} \log\left\{\frac{f(z_i : \boldsymbol{\theta}_0)}{f(z_i : \boldsymbol{\theta})}\right\} f(z_i : \boldsymbol{\theta}_0)\, dz_i = \sum_{i=1}^n K(\boldsymbol{\theta} : \boldsymbol{\theta}_0) = nK(\boldsymbol{\theta} : \boldsymbol{\theta}_0)$$

and we will therefore investigate the properties of $K(\boldsymbol{\theta} : \boldsymbol{\theta}_0)$.

Remember that log(x) is concave, and -log(x) is convex.

Notation

The support function is the loglikelihood

The score is the 1st derivative of the loglikelihood

The hessian is the 2nd derivative of the loglikelihood

The fisher information is the expected value of the outer product of the score

Under certain conditions, the fisher information is also negative expected value of the hessian

Conditions

MLE estimator itself *exists*.

In practical terms, need to check that:

1. loglik is continuous and differentiable wrt theta
2. loglik is bounded (finite)

3. limits of loglik, and check that this crosses over 0 (implies a turning point, and hence existence)

Alternatively, use the second derivative directly:

Proposition 5.6 *If Θ is a compact(closed and bounded) subset of $\mathbb{R}^k$ and the support is a continuous function of $\boldsymbol{\theta}$ then the MLE exists. Moreover, if the support function is also differentiable in $\boldsymbol{\theta}$ and the MLE $\widehat{\boldsymbol{\theta}}$ is interior to Θ then*

$$S\left(\widehat{\boldsymbol{\theta}} : z_1, \ldots, z_n\right) = S\left(\boldsymbol{\theta} : z_1, \ldots, z_n\right)\big|_{\boldsymbol{\theta}=\widehat{\boldsymbol{\theta}}} = \mathbf{0}$$

If the parameter space is convex, and loglik is concave (negative 2nd derivative), then the MLE *exists* and is *unique* (note that uniqueness impliex existence).

Proposition 5.7 *If the parameter space Θ is convex and $\log L(\boldsymbol{\theta} : z_1, \ldots, z_n)$ is a concave function of $\boldsymbol{\theta}$ then the MLE $\widehat{\boldsymbol{\theta}}$ is unique.*

Statistical Properties of MLE

Theorem 5.8 *Let $f(z : \theta)$ be a family of p.d.f.'s on S_Z where $\theta \in \Theta \subseteq \mathbb{R}$ and let $T(z_1, \ldots, z_n)$ be an unbiased estimator of the parametric function $\phi(\theta)$. Then (subject to appropriate regularity conditions) $T(z_1, \ldots, z_n)$ has a variance*

$$Var\left[T\right] \geq \left[\phi'\left(\theta\right)\right]^2 \frac{1}{nE\left[\left(\frac{\partial \log f(z:\theta)}{\partial \theta}\right)^2\right]}\Bigg|_{\theta=\theta_0} = \frac{\left[\phi'\left(\theta_0\right)\right]^2}{ni(\theta_0)}$$

where $\phi'(\theta) = \frac{\partial \phi(\theta)}{\partial \theta}$ and θ_0 denotes the true parameter value.

Easy way to check that some *unbiased* estimator reaches the CRLB.

Corollary 5.9 *There exists an unbiased estimator $T(z_1, \ldots, z_n)$ of $\phi(\theta)$ whose variance achieves the Cramer-Rao lower bound if and only if the score $S(\theta : z_1, \ldots, z_n)$ can be expressed as*

$$S\left(\theta : z_1, \ldots, z_n\right) = K\left(\theta\right)\left\{T\left(z_1, \ldots, z_n\right) - \phi\left(\theta\right)\right\}$$

where $K(\theta) = (n \times i(\theta))/\phi'(\theta)$.

Practically, in the case of MLE, K(theta) can actually be *any* arbitrary function of theta, as noted by Casella and Berger, pg 364.

The corollary leads to the conclusion that the C.R.L.B. is achieved by the MLE of $\phi(\theta)$ because by definition

$$S\left(\widehat{\theta} : z_1, \ldots, z_n\right) = 0 \implies K\left(\widehat{\theta}\right)\left\{T(z_1, \ldots, z_n) - \phi\left(\widehat{\theta}\right)\right\} = 0$$

and $K\left(\widehat{\theta}\right) \neq 0$ by assumption. Thus

$$T(z_1, \ldots, z_n) = \phi\left(\widehat{\theta}\right)$$

i.e. $T(z_1, \ldots, z_n)$ is the MLE of the parametric function $\phi(\theta)$.

Example: Consider the classical linear model in which $y_i = \mathbf{x}_i'\boldsymbol{\beta} + \varepsilon_i$ and the conditional p.d.f. $y_i \sim N(\mathbf{x}_i'\boldsymbol{\beta}, \sigma_\varepsilon^2)$, or equivalently $y_i - \mathbf{x}_i'\boldsymbol{\beta}$ are i.i.d $N(0, \sigma_\varepsilon^2)$, with $\mathbf{x}$ exogenous. Then $\boldsymbol{\theta} = (\boldsymbol{\beta}', \sigma_\varepsilon^2)' \in \Theta = \mathbb{R}^k \times \mathbb{R}^+$, the log likelihood (support)

$$\log L(\boldsymbol{\theta} : y_1, \ldots, y_n) = -\frac{1}{2}\left\{n \log 2\pi + n \log \sigma_\varepsilon^2 + \frac{\sum_{i=1}^n (y_i - \mathbf{x}_i'\boldsymbol{\beta})^2}{\sigma_\varepsilon^2}\right\},$$

and the score vector

$$\frac{\partial \log L(\boldsymbol{\theta} : y_1, \ldots, y_n)}{\partial \boldsymbol{\theta}} = \begin{bmatrix} \frac{\partial \log L}{\partial \boldsymbol{\beta}} \\ \frac{\partial \log L}{\partial \sigma_\varepsilon^2} \end{bmatrix}$$

where

$$\frac{\partial \log L}{\partial \boldsymbol{\beta}} = \frac{\sum_{i=1}^n \mathbf{x}_i (y_i - \mathbf{x}_i'\boldsymbol{\beta})}{\sigma_\varepsilon^2}$$

OLS MLE

and

$$\frac{\partial \log L}{\partial \sigma_\varepsilon^2} = -\frac{n}{2\sigma_\varepsilon^2} + \frac{\sum_{i=1}^n (y_i - \mathbf{x}_i'\boldsymbol{\beta})^2}{2(\sigma_\varepsilon^2)^2}.$$

Solving the score equations gives the MLE

$$\left.\begin{bmatrix} \frac{\partial L}{\partial \boldsymbol{\beta}} \\ \frac{\partial L}{\partial \sigma_\varepsilon^2} \end{bmatrix}\right|_{\substack{\boldsymbol{\beta}=\widehat{\boldsymbol{\beta}} \\ \sigma_\varepsilon^2=\widehat{\sigma}_\varepsilon^2}} = \begin{bmatrix} \mathbf{0} \\ 0 \end{bmatrix} \Longrightarrow \begin{bmatrix} \widehat{\boldsymbol{\beta}} = & (\sum_{i=1}^n \mathbf{x}_i\mathbf{x}_i')^{-1}(\sum_{i=1}^n \mathbf{x}_i y_i) = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} \\ \widehat{\sigma}_\varepsilon^2 = & \sum_{i=1}^n (y_i - \mathbf{x}_i'\widehat{\boldsymbol{\beta}})^2/n = \mathbf{e}'\mathbf{e}/n \end{bmatrix}$$

which coincides with the OLS regression solution $\mathbf{y} = \mathbf{X}\mathbf{b} + \mathbf{e}$ where $\mathbf{b} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$.

To obtain $ni(\boldsymbol{\theta})$, a $(k+1) \times (k+1)$ matrix, now differentiate $\log L(\boldsymbol{\theta} : y_1, \ldots, y_n)$ directly a second time to give

$$\begin{aligned} \frac{\partial^2 \log L}{\partial \boldsymbol{\beta}\partial \boldsymbol{\beta}'} &= -\frac{\sum_{i=1}^n \mathbf{x}_i\mathbf{x}_i'}{\sigma^2} \\ \frac{\partial^2 \log L}{\partial \boldsymbol{\beta}\partial \sigma^2} &= -\frac{\sum_{i=1}^n \mathbf{x}_i (y_i - \mathbf{x}_i'\boldsymbol{\beta})}{(\sigma^2)^2} = \left[\frac{\partial^2 \log L}{\partial \sigma^2 \partial \boldsymbol{\beta}'}\right]' \\ \frac{\partial^2 \log L}{\partial (\sigma^2)^2} &= \frac{n}{2}\frac{1}{(\sigma^2)^2} - \frac{\sum_{i=1}^n (y_i - \mathbf{x}_i'\boldsymbol{\beta})^2}{(\sigma^2)^3} \end{aligned}$$

and thus that

$$n\mathbf{i}(\boldsymbol{\theta}_0) = -E\left[\frac{\partial^2 \log L(\boldsymbol{\theta}_0 : y_1, \ldots, y_n)}{\partial \boldsymbol{\theta}\partial \boldsymbol{\theta}'}\right] = \begin{bmatrix} \frac{\sum_{i=1}^n \mathbf{x}_i\mathbf{x}_i'}{\sigma_0^2} & \mathbf{0} \\ \mathbf{0}' & \frac{n}{2\sigma_0^4} \end{bmatrix} = \begin{bmatrix} \sigma_0^{-2}(\mathbf{X}'\mathbf{X}) & \mathbf{0} \\ \mathbf{0}' & \frac{n}{2\sigma_0^4} \end{bmatrix}. \square$$

Asymptotic Properties

General distribution of MLE:

Proposition 5.14 *Given assumptions (i) to (vii) there exists a sequence of maximum likelihood estimators $\widehat{\boldsymbol{\theta}}$ such that*

$$\sqrt{n}\left(\widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0\right) \xrightarrow{\mathcal{D}} N\left(\mathbf{0}, \mathbf{i}(\boldsymbol{\theta}_0)^{-1}\right).$$

Note that the information matrix relies on *true* theta0, so instead use i(theta hat) as a consistent estimator for it.

Inference

Restrictions are denoted

Consider the general problem of testing the null hypotheses

$$H_0 : \boldsymbol{\theta} \in \Theta_0 \subseteq \Theta$$

where the subset Θ_0 is characterised by the "restrictions"

$$\Theta_0 = \{\boldsymbol{\theta} : h(\boldsymbol{\theta}) = \mathbf{0}\} \ ,$$

wherein

$$h(\boldsymbol{\theta}) = \begin{bmatrix} h_1(\boldsymbol{\theta}) \\ \vdots \\ h_r(\boldsymbol{\theta}) \end{bmatrix}$$

and $r \leq p$, r is the number of restrictions. The functions $h_1(\cdot), \ldots, h_r(\cdot)$ are assumed to be linearly independent, so there are no constants λ_i, not all of which are zero, $i = 1, \ldots, r$, such that $\sum_{i=1}^{r} \lambda_i h_i(\boldsymbol{\theta}) = \mathbf{0}$.

$$TW_n = nh(\boldsymbol{\theta})' \left[\frac{\partial h(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}'} \widehat{\mathbf{i}(\boldsymbol{\theta})}^{-1} \frac{\partial h(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} \right]^{-1} h(\boldsymbol{\theta}) \Bigg|_{\boldsymbol{\theta}=\hat{\boldsymbol{\theta}}}$$
$$\xrightarrow{\mathcal{D}} \chi^2(r)$$

Wald Test

Proof: 1st order Taylor Expand h(theta hat) about true theta 0

$$TS_n = n^{-1} S(\widetilde{\boldsymbol{\theta}})' \widehat{i(\widetilde{\boldsymbol{\theta}})}^{-1} S(\widetilde{\boldsymbol{\theta}})$$
$$= n^{-1} \widetilde{\boldsymbol{\lambda}}' \left[\frac{\partial h(\widetilde{\boldsymbol{\theta}})}{\partial \boldsymbol{\theta}'} \widehat{i(\widetilde{\boldsymbol{\theta}})}^{-1} \frac{\partial h(\widetilde{\boldsymbol{\theta}})}{\partial \boldsymbol{\theta}} \right] \widetilde{\boldsymbol{\lambda}}$$
$$\xrightarrow{\mathcal{D}} \chi^2(r)$$

LM/Gradient Test

The proof in the notes has typos.

Likelihood Ratio (criterion differnce) test The easiest one!

Proposition 7.3 *Under assumptions (i)-(vii) the test of H_0 defined by the critical region $\{TLR_n \geq \chi^2_{1-\alpha}(r)\}$ where*

$$TLR_n = 2\left[\log L\left(\widehat{\boldsymbol{\theta}} : z_1, \ldots, z_n\right) - \log L\left(\widetilde{\boldsymbol{\theta}} : z_1, \ldots, z_n\right)\right],$$

where $\widehat{\boldsymbol{\theta}}$ unrestricted and $\widetilde{\boldsymbol{\theta}}$ is restricted MLE, has asymptotic level α and is consistent.

Proof is harder, requires 2nd order Taylor Series expansion.

Stochastic Order

Generalised Method of Moments

Note that Method of Moments is too easy and skipped (not too exciting).

There are p parameters, and k moment conditions

The true moment conditions are denoted by h, s.t. E(h) = 0

The sample moment conditions are denoted by m, whose sample (empirical) means correspond to E(h)

A better idea is to use all components and choose $\widehat{\boldsymbol{\theta}}$ to make the moment conditions hold as nearly as possible. Specifically the idea is to minimize the quadratic form

$$Q_n(\boldsymbol{\theta}) = \mathbf{m}(\mathbf{W}, \boldsymbol{\theta})'\mathbf{W}_n^{-1}\mathbf{m}(\mathbf{W}, \boldsymbol{\theta}),$$

where $\mathbf{W}_n$ is some $k \times k$ positive definite symmetric matrix that determines the relative weights given to the components of $\mathbf{m}(\mathbf{W}, \boldsymbol{\theta})$. This leads to the definition

$$\widehat{\boldsymbol{\theta}} = \arg\min_{\theta \in \Theta} Q_n(\boldsymbol{\theta}).$$

The first order condition for a minimum of the criterion $Q_n(\boldsymbol{\theta})$ at $\boldsymbol{\theta}$ is

$$\partial Q_n(\boldsymbol{\theta})/\partial\boldsymbol{\theta} = 2\mathbf{D}(\mathbf{W}, \boldsymbol{\theta})'\mathbf{W}_n^{-1}\mathbf{m}(\mathbf{W}, \boldsymbol{\theta}) = \mathbf{0}$$

where

$$\mathbf{D}(\mathbf{W}, \boldsymbol{\theta}) = \frac{\partial \mathbf{m}(\mathbf{W}, \boldsymbol{\theta})}{\partial \boldsymbol{\theta}'}$$

is an $k \times p$ gradient matrix.

GMM is identified if:

Lemma 10.1 *(Order Condition): If $\boldsymbol{\theta}_0$ is globally identified then $k \geq p$.*

This gives a simple necessary condition for identification, namely, the number of moment conditions must be greater than or equal to the number of parameters.

The following gives us a sufficient condition for local identification when the model is "just-identified" or "overidentified", *i.e* when $k \geq p$.

Lemma 10.2 *If $\mathbf{h}_t(\mathbf{W}_t, \boldsymbol{\theta})$ is differentiable in $\boldsymbol{\theta}$,*

$$\frac{\partial \bar{\mathbf{h}}(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}'} = \bar{\boldsymbol{\Delta}}(\boldsymbol{\theta}) = \frac{1}{n}\sum_{t=1}^{n} E\left[\frac{\partial \mathbf{h}_t(\mathbf{W}_t, \boldsymbol{\theta})}{\partial \boldsymbol{\theta}'}\right],$$

and $\bar{\boldsymbol{\Delta}}(\boldsymbol{\theta})$ has full column rank in a neighbourhood $N(\boldsymbol{\theta}_0) \subseteq \Theta$ of $\boldsymbol{\theta}_0$, then $\boldsymbol{\theta}_0$ is locally identified, that is, $\bar{\mathbf{h}}(\boldsymbol{\theta}) \neq \mathbf{0}$ for all $\boldsymbol{\theta} \in N(\boldsymbol{\theta})$, $\boldsymbol{\theta} \neq \boldsymbol{\theta}_0$.

Asymptotics

$$\sqrt{n}(\widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) \xrightarrow{\mathcal{D}} \mathfrak{N}(\mathbf{0}, \bar{\mathbf{V}}_0)$$

$$\bar{\mathbf{V}}_0 = (\bar{\boldsymbol{\Delta}}_0' \mathbf{W}_0^{-1} \bar{\boldsymbol{\Delta}}_0)^{-1} (\bar{\boldsymbol{\Delta}}_0' \mathbf{W}_0^{-1} \bar{\boldsymbol{\Sigma}}_0 \mathbf{W}_0^{-1} \bar{\boldsymbol{\Delta}}_0)(\bar{\boldsymbol{\Delta}}_0' \mathbf{W}_0^{-1} \bar{\boldsymbol{\Delta}}_0)^{-1},$$

$\bar{\boldsymbol{\Delta}}_0 = \bar{\boldsymbol{\Delta}}(\boldsymbol{\theta}_0)$ *and* $\bar{\boldsymbol{\Sigma}}_0 = \bar{\boldsymbol{\Sigma}}(\boldsymbol{\theta}_0)$.

The proof is a straightforward, by tedious Taylor Series expansion about m(theta hat)

Relative Efficiency

Lemma 10.3 *The asymptotic variance of the GMM estimator is bounded below by $(\bar{\boldsymbol{\Delta}}_0' \bar{\boldsymbol{\Sigma}}_0^{-1} \bar{\boldsymbol{\Delta}}_0)^{-1}$, that is $\bar{\mathbf{V}}_0 \geq (\bar{\boldsymbol{\Delta}}_0' \bar{\boldsymbol{\Sigma}}_0^{-1} \bar{\boldsymbol{\Delta}}_0)^{-1}$.*

Proof: Consider the difference $(\bar{\mathbf{\Delta}}_0'\bar{\mathbf{\Sigma}}_0^{-1}\bar{\mathbf{\Delta}}_0)-\bar{\mathbf{V}}_0^{-1}$. If we can show that $(\bar{\mathbf{\Delta}}_0'\bar{\mathbf{\Sigma}}_0^{-1}\bar{\mathbf{\Delta}}_0)-\bar{\mathbf{V}}_0^{-1} \geq 0$, then $(\bar{\mathbf{\Delta}}_0'\bar{\mathbf{\Sigma}}_0^{-1}\bar{\mathbf{\Delta}}_0) \geq \bar{\mathbf{V}}_0^{-1}$ and $\bar{\mathbf{V}}_0 \geq (\bar{\mathbf{\Delta}}_0'\bar{\mathbf{\Sigma}}_0^{-1}\bar{\mathbf{\Delta}}_0)^{-1}$, as required.

Let $\mathbf{A}' = \bar{\mathbf{\Delta}}_0'\bar{\mathbf{\Sigma}}_0^{-\frac{1}{2}}$ and $\mathbf{B}' = \bar{\mathbf{\Delta}}_0'\mathbf{W}_0^{-1}\bar{\mathbf{\Sigma}}_0^{\frac{1}{2}}$ where $\bar{\mathbf{\Sigma}}_0 = \bar{\mathbf{\Sigma}}_0^{\frac{1}{2}}(\bar{\mathbf{\Sigma}}_0^{\frac{1}{2}})'$ and $\bar{\mathbf{\Sigma}}_0^{\frac{1}{2}} = (\bar{\mathbf{\Sigma}}_0^{\frac{1}{2}})'$. Then

$$\begin{aligned}\bar{\mathbf{V}}_0^{-1} &= (\bar{\mathbf{\Delta}}_0'\mathbf{W}_0^{-1}\bar{\mathbf{\Delta}}_0)(\bar{\mathbf{\Delta}}_0'\mathbf{W}_0^{-1}\bar{\mathbf{\Sigma}}_0\mathbf{W}_0^{-1}\bar{\mathbf{\Delta}}_0)^{-1}(\bar{\mathbf{\Delta}}_0'\mathbf{W}_0^{-1}\bar{\mathbf{\Delta}}_0) \\ &= \mathbf{A}'\mathbf{B}(\mathbf{B}'\mathbf{B})^{-1}\mathbf{B}'\mathbf{A}\end{aligned}$$

and

$$\bar{\mathbf{\Delta}}_0'\bar{\mathbf{\Sigma}}_0^{-1}\bar{\mathbf{\Delta}}_0 = \mathbf{A}'\mathbf{A}\,.$$

The difference $(\bar{\mathbf{\Delta}}_0'\bar{\mathbf{\Sigma}}_0^{-1}\bar{\mathbf{\Delta}}_0) - \bar{\mathbf{V}}_0^{-1}$ becomes

$$\mathbf{A}'\mathbf{A} - \mathbf{A}'\mathbf{B}(\mathbf{B}'\mathbf{B})^{-1}\mathbf{B}'\mathbf{A} = \mathbf{A}'(\mathbf{I} - \mathbf{B}(\mathbf{B}'\mathbf{B})^{-1}\mathbf{B}')\mathbf{A} \geq 0$$

because $\mathbf{I} - \mathbf{B}(\mathbf{B}'\mathbf{B})^{-1}\mathbf{B}'$ is a projection matrix and therefore positive semi-definite. Thus $\mathbf{A}'\mathbf{A} \geq \mathbf{A}'\mathbf{B}(\mathbf{B}'\mathbf{B})^{-1}\mathbf{B}'\mathbf{A}$, implying that $(\mathbf{A}'\mathbf{A})^{-1} \leq (\mathbf{A}'\mathbf{B})^{-1}(\mathbf{B}'\mathbf{B})(\mathbf{B}'\mathbf{A})^{-1}$.

Weighting matrix

A an efficient weighting matrix needs to chosen up to a proportionality:

Theorem 10.4 *Under assumptions M1-M5 the GMM estimator $\widehat{\boldsymbol{\theta}}$ associated with $\mathbf{W}_n$ is asymptotically efficient relative to all other GMM estimators if and only if there exists a $p \times p$ nonsingular matrix $\mathbf{C}$ such that*

$$\mathbf{W}_0^{-1}\bar{\mathbf{\Delta}}_0 = \bar{\mathbf{\Sigma}}_0^{-1}\bar{\mathbf{\Delta}}_0\mathbf{C}\,.$$

In practice, is it easier to directly work out the variance of the moment conditions and get the standard efficient weighting matrix.

However, the previous result is useful, as this allows us to be sloppy with constants (e.g. missing Ts).

OLS Worked example

In the GMM setting we therefore have

$$\mathbf{m}(\mathbf{W}, \boldsymbol{\beta}) = \frac{1}{T}\sum_{t=1}^{T} \mathbf{x}_t(y_t - \mathbf{x}_t'\boldsymbol{\beta}) = \frac{1}{T}\mathbf{X}'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})\,.$$

For any weighting matrix $\mathbf{W}_T$ the foc condition for the minimization of

$$Q_T(\boldsymbol{\beta}) = \left(\frac{(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})'\mathbf{X}}{T}\right)\mathbf{W}_T^{-1}\left(\frac{\mathbf{X}'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})}{T}\right)$$

is

$$2T^{-2}(\mathbf{X}'\mathbf{X})\mathbf{W}_T^{-1}\mathbf{X}'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) = \mathbf{0}$$

since

$$\mathbf{D}(\mathbf{W}_t, \boldsymbol{\beta}) = \frac{\partial \mathbf{m}(\mathbf{W}, \boldsymbol{\beta})}{\partial \boldsymbol{\beta}'} = -\frac{1}{T}\sum_{t=1}^{T}\mathbf{x}_t\mathbf{x}_t' = -\left(\frac{\mathbf{X}'\mathbf{X}}{T}\right)\,.$$

IV Regression Worked Example. Note that sometimes Don is sloppy and specifies different T proportionalities.

We now implement GMM using the moment condition $E[\mathbf{h}_t(\mathbf{W}_t, \boldsymbol{\beta})] = \mathbf{0}$ where $\mathbf{h}_t(\mathbf{W}_t, \boldsymbol{\beta}) = \mathbf{z}_t(y_t - \mathbf{x}_t'\boldsymbol{\beta})$, $\mathbf{W}_t' = (y_t, \mathbf{x}_t', \mathbf{z}_t')$. This has the sample counterpart

$$\mathbf{m}(\mathbf{W}_t, \boldsymbol{\beta}) = \frac{1}{T}\sum_{t=1}^{T} \mathbf{z}_t(y_t - \mathbf{x}_t'\boldsymbol{\beta}) = \frac{1}{T}\mathbf{Z}'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})\,.$$

For any weighting matrix $\mathbf{W}_T$ the foc condition for the minimization of

$$Q_T(\boldsymbol{\beta}) = \left(\frac{(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})'\mathbf{Z}}{T}\right)\mathbf{W}_T^{-1}\left(\frac{\mathbf{Z}'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})}{T}\right)$$

is

$$2T^{-2}(\mathbf{Z}'\mathbf{X})'\mathbf{W}_T^{-1}\mathbf{Z}'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) = \mathbf{0}$$

since

$$\mathbf{D}(\mathbf{W}_t, \boldsymbol{\beta}) = \frac{\partial \mathbf{m}(\mathbf{W}, \boldsymbol{\beta})}{\partial \boldsymbol{\beta}'} = -\frac{1}{T}\sum_{t=1}^{T}\mathbf{z}_t\mathbf{x}_t' = -\left(\frac{\mathbf{Z}'\mathbf{X}}{T}\right)\,.$$

Assuming that $E[\varepsilon_t \varepsilon_s | \mathbf{z}_t, \mathbf{z}_s] = \sigma_\varepsilon^2$ when $t = s$ and is zero otherwise, we get

$$\bar{\mathbf{\Sigma}} = TE[\mathbf{m}(\mathbf{W}, \boldsymbol{\beta}_0)\mathbf{m}(\mathbf{W}, \boldsymbol{\beta}_0)'] = \sigma_\varepsilon^2 T^{-1} \sum_{t=1}^{T} E[\mathbf{z}_t \mathbf{z}_t'] \,.$$

If the instruments satisfy sufficient regularity to ensure that

$$\text{plim} \left\| \frac{\mathbf{Z}'\mathbf{Z}}{T} - \frac{1}{T} \sum_{t=1}^{T} E[\mathbf{z}_t \mathbf{z}_t'] \right\| = 0$$

then an optimal weighting matrix is given by $\mathbf{W}_T = T^{-1}(\mathbf{Z}'\mathbf{Z})$. With this choice of weighting matrix the foc becomes equivalent to

$$(\mathbf{X}'\mathbf{Z})(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) = \mathbf{0},$$

MLE Worked Example

Inference

Wald Test

LM/Gradient Test

Criterion Difference Test

Notation

Inference

Extremum Estimators

Censoring and Truncation

Truncated

A *truncated* distribution is where any observations exceeding some threshold are not observed at all.

Censored

A *censored* distribution is where any observations exceeding some threshold are simply recorded as that threshold, but its actual value is unknown (censored).

Generalised Linear Models (GLMs)